

MACHINE LEARNING LAB MANUAL



AMRITA

VISHWA VIDYAPEETHAM

DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

Department of Computer Science and Engineering

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Aim : To perform exploratory data analysis (EDA) on a student performance dataset using Pandas and Matplotlib.

Dataset : Create a CSV file named student_data.csv with the following data:

RollNo	Name	Gender	Attendance	InternalMarks	ExternalMarks
101	Ravi	M	85	32	55
102	Priya	F	92	38	58
103	Kiran	M	78	28	45
104	Anu	F	95	40	60
105	Sita	F	88	35	54
106	Raj	M	70	25	40
107	Divya	F	90	37	57
108	Arun	M	82	30	48
109	Keerthi	F	96	39	59
110	Naveen	M	75	27	44

Tasks

Task A: Dataset Loading and Inspection

- Import the necessary Python libraries.
- Load the dataset into a Pandas DataFrame.
- Display the first five records.
- Display the last five records.
- Find the total number of rows and columns.
- Display the names of all columns.
- Display the data type of each column.
- Generate dataset information using info().

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("student_data.csv")
print(df.head())
```

```
print(df.tail())  
print(df.shape)  
print(df.columns)  
print (df.dtypes)  
print (df.info())
```

Output:

	RollNo	Name	Gender	Attendance	InternalMarks	ExternalMarks
0	101	Ravi	M	85	32	55
1	102	Priya	F	92	38	58
2	103	Kiran	M	78	28	45
3	104	Anu	F	95	40	60
4	105	Sita	F	88	35	54

	RollNo	Name	Gender	Attendance	InternalMarks	ExternalMarks
5	106	Raj	M	70	25	40
6	107	Divya	F	90	37	57
7	108	Arun	M	82	30	48
8	109	Keerthi	F	96	39	59
9	110	Naveen	M	75	27	44

```
(10, 6)
```

```
[Done] exited with code=0 in 1.391 seconds
```

```
Index(['RollNo', 'Name', 'Gender', 'Attendance', 'InternalMarks',  
      'ExternalMarks'],  
      dtype='str')
```

85.1

33.1

```
RollNo      int64  
Name        str  
Gender      str  
Attendance  int64  
InternalMarks int64  
ExternalMarks int64  
dtype: object  
85.1  
33.1
```

```
In [10]: class 'pandas.DataFrame'  
RangeIndex: 10 entries, 0 to 9  
Data columns (total 6 columns):  
#   Column      Non-Null Count  Dtype  
---  ---  
0   RollNo      10 non-null    int64  
1   Name        10 non-null    str  
2   Gender      10 non-null    str  
3   Attendance  10 non-null    int64  
4   InternalMarks 10 non-null    int64  
5   ExternalMarks 10 non-null    int64  
dtypes: int64(4), str(2)  
memory usage: 612.0 bytes  
Done  
85.1  
33.1  
[Done] exited with code=0 in 1.426 seconds
```

Task B: Statistical Analysis

- Calculate the mean attendance of all students.
- Calculate the average Internal Marks.
- Calculate the average External Marks.
- Find the maximum Internal Marks.

- Find the minimum Internal Marks.
- Find the maximum External Marks.
- Find the minimum External Marks.
- Calculate the standard deviation of Internal Marks.
- Generate summary statistics using describe().

Code:

```
print(df["Attendance"].mean())
print(df["InternalMarks"].mean())
print(df["ExternalMarks"].mean())
print(df["InternalMarks"].max())
print(df["InternalMarks"].min())
print(df["ExternalMarks"].max())
print(df["ExternalMarks"].min())
print(df["ExternalMarks"].std())
print(df["InternalMarks"].std())
print(df.describe())
```

Output:

85.1				
33.1				
52.0				
40				
25				
60				
40				
7.1492035298424055				
5.425249610233001				
	RollNo	Attendance	InternalMarks	ExternalMarks
count	10.00000	10.000000	10.00000	10.000000
mean	105.50000	85.100000	33.10000	52.000000
std	3.02765	8.736259	5.42525	7.149204
min	101.00000	70.000000	25.00000	40.000000
25%	103.25000	79.000000	28.50000	45.750000
50%	105.50000	86.500000	33.50000	54.500000
75%	107.75000	91.500000	37.75000	57.750000
max	110.00000	96.000000	40.00000	60.000000

Task C: Data Manipulation

- Create a new column named TotalMarks.
- TotalMarks=InternalMarks+ExternalMarks
- Create a new column named Percentage.

1. Percentage=TotalMarks/100×100 • Display the updated DataFrame.

Code:

```
df["TotalMarks"] = df["InternalMarks"] + df["ExternalMarks"]
```

```
df["Percentage"] = (df["TotalMarks"] / 100) * 100
```

Output:

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

[Running]

python -u "c:\Users\Sreedurga\OneDrive - Amrita Vishwa Vidyapeetham\Desktop\ML\task1.py"

	RollNo	Name	Gender	...	ExternalMarks	TotalMarks	Percentage
0	101	Ravi	M	...	55	87	87.0
1	102	Priya	F	...	58	96	96.0
2	103	Kiran	M	...	45	73	73.0
3	104	Anu	F	...	60	100	100.0
4	105	Sita	F	...	54	89	89.0
5	106	Raj	M	...	40	65	65.0
6	107	Divya	F	...	57	94	94.0
7	108	Arun	M	...	48	78	78.0
8	109	Keerthi	F	...	59	98	98.0
9	110	Naveen	M	...	44	71	71.0

[10 rows x 8 columns]

	RollNo	Name	Gender	...	ExternalMarks	TotalMarks	Percentage
1	102	Priya	F	...	58	96	96.0
3	104	Anu	F	...	60	100	100.0
8	109	Keerthi	F	...	59	98	98.0

Task D: Filtering and Searching

- Display the details of students whose attendance is greater than 90%.
- Display the details of students whose TotalMarks exceed 90.
- Display the details of female students.
- Display the details of male students.
- Display students whose ExternalMarks are less than 50.

Code:

```
print(df[df["Attendance"]>90])
```

```
print(df[df["TotalMarks"]>90])
```

```
print(df[df["Gender"] == "F"])
print(df[df["Male"] == "M"])
print(df[df["ExternalMarks"]<50])
```

Output:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

[10 rows x 8 columns]

	RollNo	Name	Gender	...	ExternalMarks	TotalMarks	Percentage
1	102	Priya	F	...	58	96	96.0
3	104	Anu	F	...	60	100	100.0
8	109	Keerthi	F	...	59	98	98.0

[3 rows x 8 columns]

	RollNo	Name	Gender	...	ExternalMarks	TotalMarks	Percentage
1	102	Priya	F	...	58	96	96.0
3	104	Anu	F	...	60	100	100.0
6	107	Divya	F	...	57	94	94.0
8	109	Keerthi	F	...	59	98	98.0

[4 rows x 8 columns]

	RollNo	Name	Gender	...	ExternalMarks	TotalMarks	Percentage
1	102	Priya	F	...	58	96	96.0
3	104	Anu	F	...	60	100	100.0
4	105	Sita	F	...	54	89	89.0
6	107	Divya	F	...	57	94	94.0
8	109	Keerthi	F	...	59	98	98.0

[5 rows x 8 columns]

	RollNo	Name	Gender	...	ExternalMarks	TotalMarks	Percentage
0	101	Ravi	M	...	55	87	87.0
2	103	Kiran	M	...	45	73	73.0
5	106	Raj	M	...	40	65	65.0
7	108	Arun	M	...	48	78	78.0
9	110	Naveen	M	...	44	71	71.0

[5 rows x 8 columns]

	RollNo	Name	Gender	...	ExternalMarks	TotalMarks	Percentage
2	103	Kiran	M	...	45	73	73.0
5	106	Raj	M	...	40	65	65.0
7	108	Arun	M	...	48	78	78.0
9	110	Naveen	M	...	44	71	71.0

Task E: Ranking and Comparison

- Identify the student with the highest TotalMarks.
- Identify the student with the lowest TotalMarks.
- Display the top three performers based on TotalMarks.
- Display the bottom three performers based on TotalMarks.
- Determine how many students scored above the class average.

Code:

```
print(df.loc[df["TotalMarks"].idxmax()])
```

```

print(df.loc[df["TotalMarks"].idxmin()])

print(df.nlargest(3, "TotalMarks"))

print(df.nsmallest(3, "TotalMarks"))

average = df["TotalMarks"].mean()

print("\nStudents Above Class Average")

print(df[df["TotalMarks"] > average])

print("Number of Students:", len(df[df["TotalMarks"] > average]))

```

Output:

```

RollNo      104
Name        Anu
Gender       F
Attendance   95
InternalMarks 40
ExternalMarks 60
TotalMarks   100
Percentage   100.0
Name: 3, dtype: object
RollNo      106
Name        Raj
Gender       M
Attendance   70
InternalMarks 25
ExternalMarks 40
TotalMarks   65
Percentage   65.0
Name: 5, dtype: object

```

	RollNo	Name	Gender	...	ExternalMarks	TotalMarks	Percentage
3	104	Anu	F	...	60	100	100.0
8	109	Keerthi	F	...	59	98	98.0
1	102	Priya	F	...	58	96	96.0

```

[3 rows x 8 columns]

```

	RollNo	Name	Gender	...	ExternalMarks	TotalMarks	Percentage
5	106	Raj	M	...	40	65	65.0
9	110	Naveen	M	...	44	71	71.0
2	103	Kiran	M	...	45	73	73.0

```

[3 rows x 8 columns]

```

Students Above Class Average

	RollNo	Name	Gender	...	ExternalMarks	TotalMarks	Percentage
0	101	Ravi	M	...	55	87	87.0
1	102	Priya	F	...	58	96	96.0
3	104	Anu	F	...	60	100	100.0
4	105	Sita	F	...	54	89	89.0
6	107	Divya	F	...	57	94	94.0
8	109	Keerthi	F	...	59	98	98.0

```

[6 rows x 8 columns]
Number of Students: 6

[Done] exited with code=0 in 1.599 seconds

```

Task F: Group-wise Analysis

- Calculate the average TotalMarks of male students.
- Calculate the average TotalMarks of female students.
- Compare the average attendance of male and female students.

Code:

```
print(df[df["Gender"] == "M"]["TotalMarks"].mean())
print(df[df["Gender"] == "F"]["TotalMarks"].mean())
print(df.groupby("Gender")["Attendance"].mean())
```

Output:

```
74.8
95.4
Gender
F    92.2
M    78.0
Name: Attendance, dtype: float64
[Done] exited with code=0 in 1.487 seconds
```

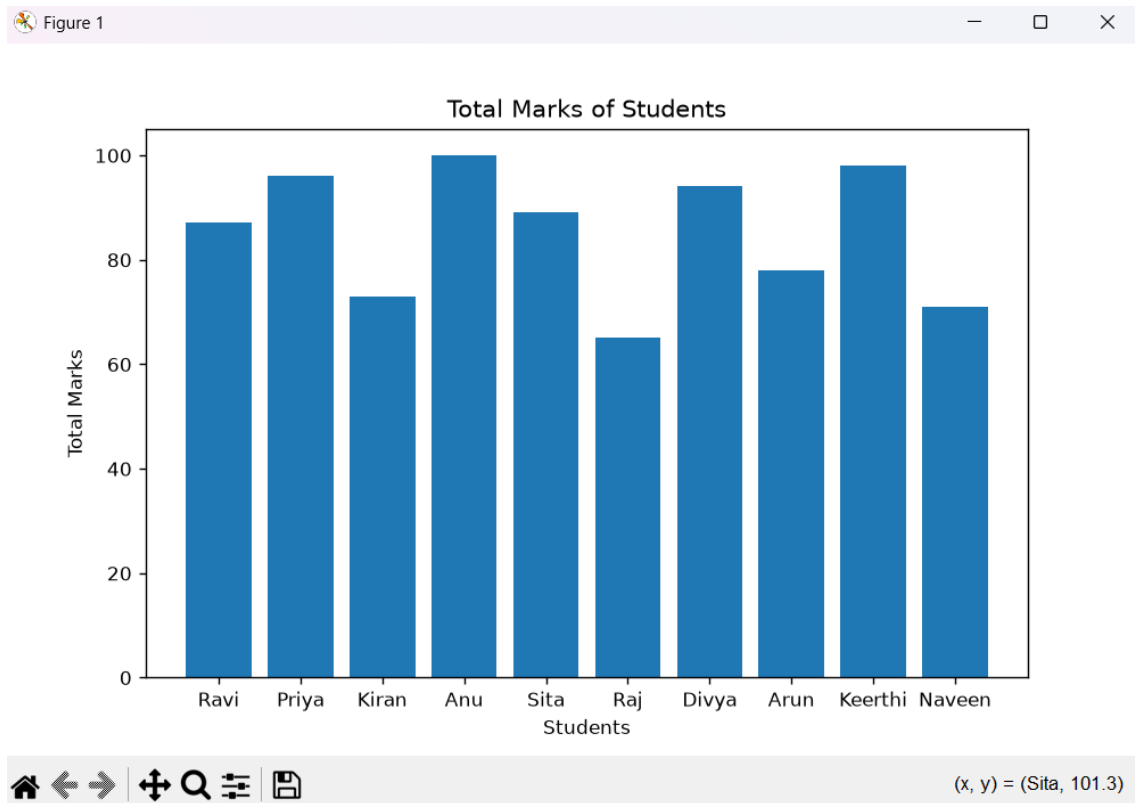
Task G: Data Visualization

- Plot a bar chart showing TotalMarks of all students.
- Plot a histogram of Attendance.
- Plot a pie chart showing the distribution of male and female students.
- Plot a scatter plot between Attendance and TotalMarks.
- Interpret the relationship between Attendance and TotalMarks.

Bar Chart Code:

```
plt.figure(figsize=(8,5))
plt.bar(df["Name"], df["TotalMarks"])
plt.title("Total Marks of Students")
plt.xlabel("Students")
plt.ylabel("Total Marks")
plt.show()
```

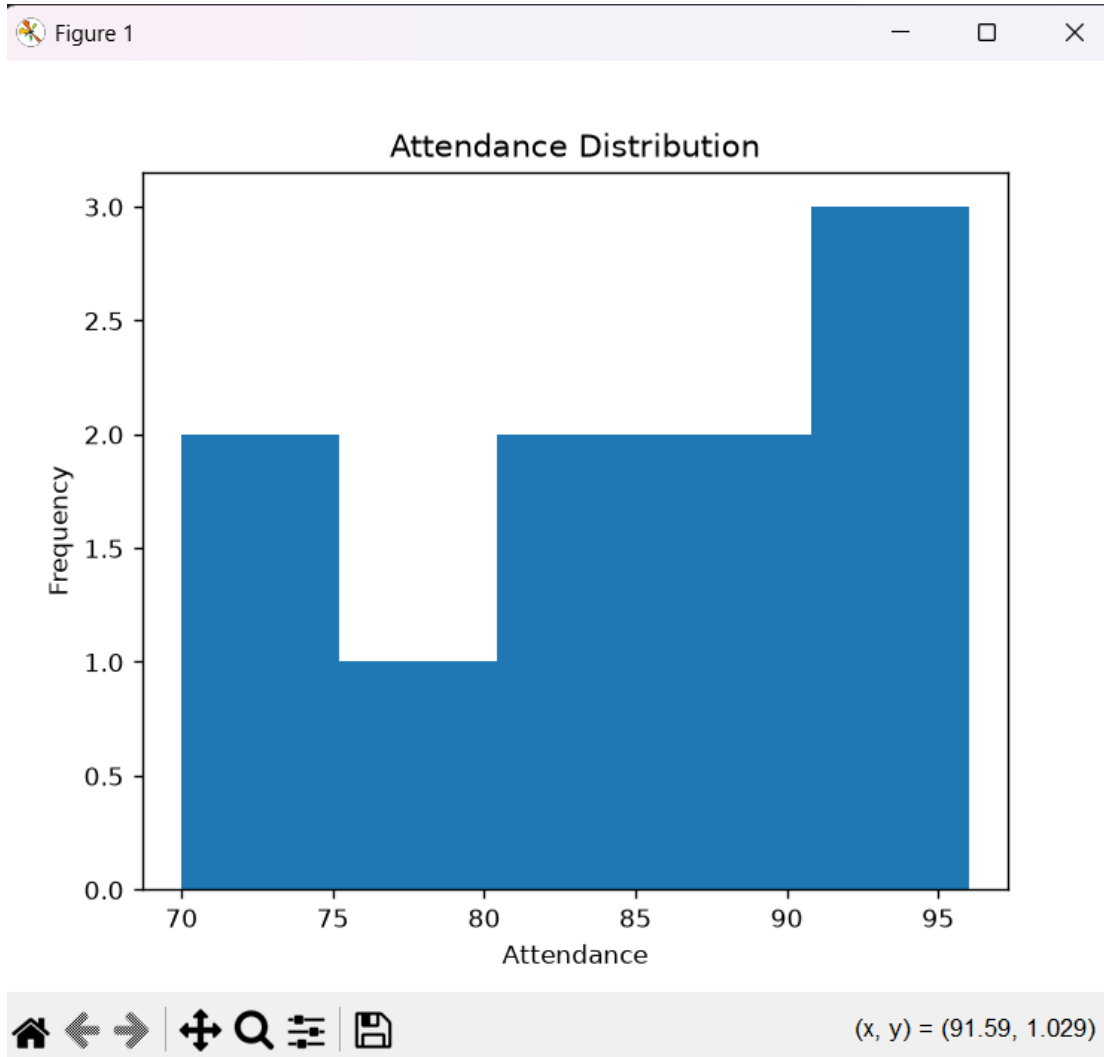
Output:



Histogram Code:

```
plt.figure(figsize=(6,5))
plt.hist(df["Attendance"], bins=5)
plt.title("Attendance Distribution")
plt.xlabel("Attendance")
plt.ylabel("Frequency")
plt.show()
```

Output:



Piechart Code:

```
gender = df["Gender"].value_counts()
```

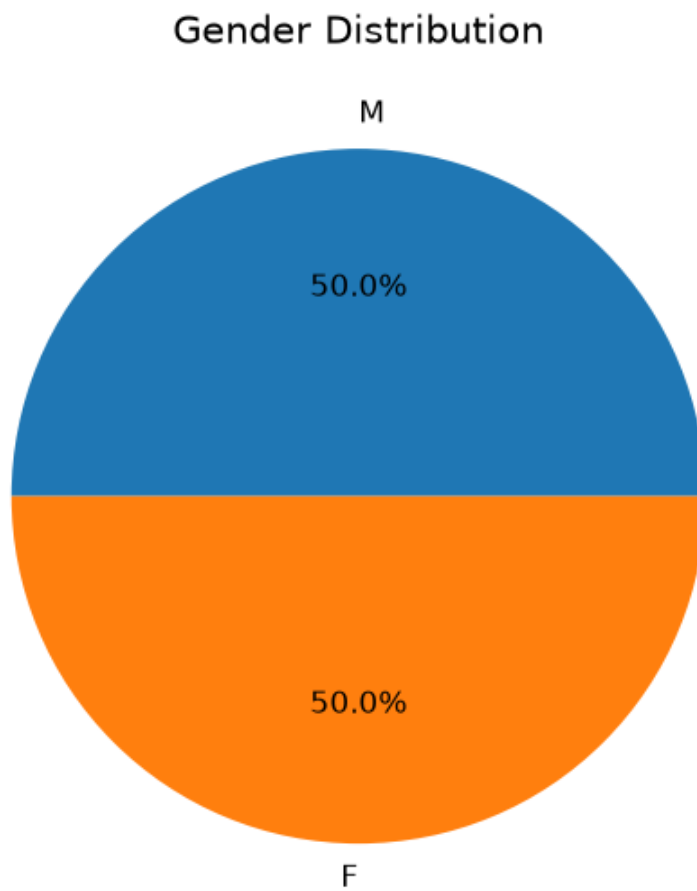
```
plt.figure(figsize=(5,5))
```

```
plt.pie(gender, labels=gender.index, autopct="%1.1f%%")
```

```
plt.title("Gender Distribution")
```

```
plt.show()
```

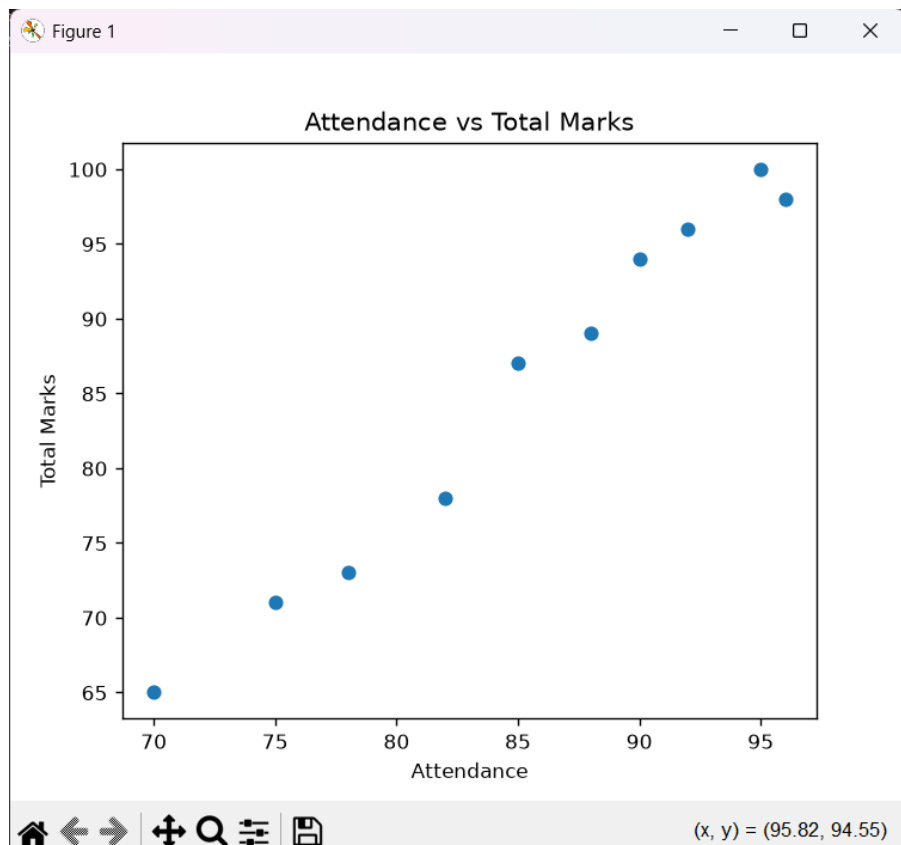
Output:



Scatter plot code:

```
plt.figure(figsize=(6,5))  
plt.scatter(df["Attendance"], df["TotalMarks"])  
plt.title("Attendance vs Total Marks")  
plt.xlabel("Attendance")  
plt.ylabel("Total Marks")  
plt.show()
```

Output:



Students with higher attendance generally scored higher TotalMarks. This indicates a positive relationship between attendance and academic performance.

Task H: Observations

- Write any three observations based on the analysis performed.
- State whether higher attendance appears to be associated with better academic performance.

Observations:

- Female students have a higher average attendance than male students.
- Anu scored the highest TotalMarks (100).
- Students with attendance above 90% generally achieved higher marks.

Exploratory Data Analysis (EDA) was successfully performed using Pandas and Matplotlib. Statistical analysis, filtering, ranking, grouping, and visualization of the student dataset were completed successfully